NISHAANTH S Quick Revision

Slides 7–12; target approximately 4.5 minutes.

18 features; TF-IDF; scaling; three models; validation selection; metrics; global explanation limits.

200 synthetic CSV rows; 40 duplicate rows removed; 160 unique examples (80 human-style, 80 AI-style); 20 synthetic problem groups; 96 training, 32 validation, 32 test samples.

Models: Logistic Regression, Decision Tree, Random Forest. Selected: Logistic Regression. Test metrics: all four 1.000; confusion matrix [[16,0],[0,16]]. Both labels are synthetic style categories written into a generator, not verified human or LLM provenance. The 20 group IDs vary a few filtering templates and do not represent 20 independent algorithmic tasks. Perfect scores describe template separation only.

How does tokenization work?
Python tokenize yields lexical tokens. The numeric token statistics include names, operators and literals while excluding comments and layout.

Does nesting depth mean raw AST depth?
No. It increments selected control constructs and function definitions; expression layers alone do not increment it.

What is the smoothed IDF formula?
1+ln((1+N)/(1+df)). The extra ones avoid zero divisions and follow TfidfVectorizer defaults.

Why did you choose XGBoost?
We did not implement it. It belongs to the synopsis and future scope; the actual project uses three scikit-learn baselines.

Demonstration recovery
Start backend with .venv\Scripts\python.exe -m uvicorn backend.main:app and frontend with npm run dev --prefix frontend. If startup fails, show the captured real screenshots and experiment CSV. Never invent a live result.